Mathematics or the Verifier? A Null Result for One Verifier-Filtered Self-Training Recipe on a 1.5B Model

Tan Sang Nguyen[1] and Duy Anh Tran[1]

Ton Duc Thang University, Ho Chi Minh City, Vietnam
52300057@student.tdtu.edu.vn, 52300006@student.tdtu.edu.vn

Abstract. Self-training and reinforcement learning with verifiable rewards report large gains for small language models, but a rising score does not show that a model learned mathematics rather than what its verifier accepts. We study the cheapest form of the loop, generate–verify–select–train with offline LoRA SFT, on Qwen2.5-1.5B-Instruct with the rule-based verifier `math-verify`, on held-out MATH-500 and GSM8K exams (150 problems; 500 in a final round) at four learning rates on MATH-500 and two on GSM8K. No pass@1 or pass@k gain has a problem-level bootstrap interval excluding zero; in the final round, with a revised (not validated) recipe, pass@1 falls 3.2–4.8 points in each of three seeds (mean -4.1; two of three intervals exclude zero), about the size of the fall in the `\boxed` share that both graders read. The design detects only gains of about 10 points of pass@1 (6 in the final round; 80% power, intervals without seed variance, so likely more). A positive gate fails: training on reference solutions lowers pass@1 by 32.2 points, which we did not diagnose (a pipeline defect would also produce it), so the null concerns this recipe only and cannot show that self-training fails. Accuracies are graded by the verifier that selects the data (the final round also by a string-match grader), so neither the mathematics nor the verifier question is answered. As a post hoc audit we track the share of the loop's own correct solutions that a storage defect in our pipeline rejects (about one in six). It falls by 1.5–2.6 points with intervals reaching zero; this fall is tied to the falling `\boxed` share by an identity, so it is no evidence of adaptation. Two injected-bias runs, one whose rule is visible in the gold answer, show no selective avoidance but are not valid positive controls (one cannot elicit avoidance by construction, the other is a single run without r_t), so the audit is uninformative about adaptation. Two controls, patching a rarer defect and removing the filter, show no consistent difference (no filter is lower on pass@k, at 3–4× the dose) but are dose-confounded or too small to detect one. Scores before the final round use the verifier with the defect.

Keywords: self-training · self-improvement · verifier bias · mathematical reasoning · small language models · verifiable rewards · null result.

1 Introduction

A common recipe for improving the mathematical reasoning of a small language model (SLM) *without human annotation* is to let the model generate its own solutions, keep those an automated verifier judges correct, and fine-tune on them, for several rounds. We call this the *Generate–Verify–Train* (GVT) loop; it underlies STaR [19], ReST [12], ReST-EM [11] and rejection-sampling fine-tuning [17]. But the verifier never reads the reasoning: a rule-based verifier such as `math-verify`[1] only extracts a final answer and compares it with the gold, and it rejects some correct answers written in forms it does not parse (Section 3.2). The training signal is "was the final string accepted", not "were the steps right".

The same gap runs through reinforcement learning with verifiable rewards (RLVR), which reports large gains at this scale [14,3,20], though only 8.6 of one study's 37.6 points lie beyond format correction. Shao et al. [9] find that GRPO with *random* rewards raises Qwen2.5-Math-7B by 21.4 points on MATH-500, against 29.1 with ground-truth rewards. A rising score therefore does not by itself show that a model has learned to solve more problems, rather than to surface behaviour it already had or to produce what its reward accepts. In RLVR, per-step false-negative rates on the policy's own rollouts have been tracked [4,16]; we study the question in a cheap and common form of the loop, offline rejection-sampling SFT, where a false reject removes a training example (and, if bound to a problem, the problem), and apply a per-round false-reject measurement there, post hoc, for one defect of our pipeline.

We ask: *after several GVT rounds, is the model better at mathematics, or has it only learned to write what the verifier accepts?* We fixed four measures in advance: (i) pass@1; (ii) pass@k; (iii) output style, chiefly the share of generations containing `\boxed`; and (iv) per-problem transitions (problems *added* and *lost* between rounds). The pre-set reading rule was that "learning the verifier's style" requires pass@1 to rise, pass@k to stay flat and style to converge at once. In hindsight the rule is one-sided: it can only fire when accuracy rises (Section 5.1). The higher learning rates, the controls, the round count and k at those rates, sample accuracy as primary measure, the termination measures and the re-scoring for a second verifier defect, and the final 500-problem round (Section 4.6) were all added after seeing earlier runs; all are post hoc.

We detect no improvement at any learning rate, though gains below about 10 points of pass@1 or 3.5–5.5 of sample accuracy could have been missed (intervals omit training-seed variance), and no arm of this recipe improves, so the result concerns the recipe. The second branch is untested: no run varies the frequent verifier defect, scores are graded by the selecting verifier (one string-match check on the final round), and the post hoc audit of Section 5.3 is uninformative about adaptation. What appears, most clearly at the highest rate, is lower accuracy on the solutions the model does finish, though on one seed the cap share carries more of it (Section 4.5). A sharp pooled decline is absent in the isolated runs at

[1] https://github.com/huggingface/math-verify, version 0.9.0 with antlr4 4.13.2 in this work.

10^{-5}, which train on about 4.5× fewer examples (Section 4.1); dose and pooling are confounded.

Contributions.

- For a 1.5B model, LoRA SFT and one rule-based verifier, no gain is detectable on a held-out exam at any learning rate we tried (a coarse grid that also changes k and rounds), with bootstrap intervals and the smallest effects the design could detect. No arm improves, not even training on reference solutions, so this is a null for the recipe, not a statement about self-training.
- A post hoc per-round false-reject audit of one verifier defect (the RLVR tracking [4,16] applied to offline SFT; not a novel method). The rejected share falls by 1.5–2.6 points with intervals reaching zero, tied to the falling `\boxed` share by an identity; the audit is uninformative about adaptation (Section 5.3).
- A decomposition of an accuracy decline under a fixed token cap into a finished-accuracy part and a cap-share part: at the highest rate, finished generations lose 5.5–10.6 points against the untrained model on the same problem mix (A, B and no-filter runs; train-on-gold loses about 26), and the cap-hit share rises only at that rate.

2 Related Work

Self-training and its limits. STaR [19] introduced the generate–filter–retrain loop, and ReST [12], ReST-EM [11] and RFT [17] formalise it; ReST-EM, which like us restarts from the base model each iteration, reports diminishing gains after the first round. Yue et al. [18] show that RL raises pass@1 while the base model wins at large pass@k, and Teacher-Free Self-Training [13] concludes that self-training "amplifies but does not compound". Training repeatedly on model-generated data erodes distribution tails [10], and a narrow selector can accelerate collapse [8].

RL with verifiable rewards at small scale. RLVR with GRPO reports large gains near our model size: one training example lifts Qwen2.5-Math-1.5B on MATH-500 from 36.0% to 73.6% [14], 7,000 examples lift DeepSeek-R1-Distill-Qwen-1.5B on AMC23 from 63% to 80% [3], and zero-RL works on open base models from 0.5B [20]. These studies improve accuracy, which our SFT loop does not; of Wang et al.'s 37.6 points only 8.6 lie beyond format correction. Shao et al. [9] show that it need not be reward learning (random rewards also help, via amplified pre-existing behaviour). We do not compete with RLVR on accuracy; the audit of Section 5.3 asks whether a loop has adapted to one verifier defect, and we apply it only to SFT.

Verifier error. TinyV [16] finds that rule-based verifiers reject many correct solutions; From Accuracy to Robustness [6] reports an average recall of 86%, and 0.95 for `math-verify` on their MATH subset. Imperfect Verifiers [1] models

fixed false-accept and false-reject rates. Egashira et al. [4] find in RLVR that systematic false negatives behave like random noise whereas false positives can cause plateau or collapse. For format-based false negatives they see the policy first unlearn the triggering behaviour. Our setting differs in paradigm (offline SFT with a fresh adapter per round) and task, and our controls cannot test their findings (Section 4.3).

Auditing self-improvement. Xu et al. [15] audit LoRA self-training of Qwen3-8B against a frozen control and identify seven measurement failures, including training-seed variance and a fixed token cap under style drift, that each invert a finding when uncontrolled. This is the study closest to ours; it is why we separate finished accuracy from termination (Section 4.5). We differ in scale (a 1.5B model), in the verifier (a rule-based one whose rejection defect we measured) and in auditing what that verifier's filtering does to the training pool, not only whether accuracy moved. Kim et al. [7] show that self-distillation with a teacher that sees the solution can cut accuracy; our train-on-gold drop resembles this, but we did not test its cause (Section 4.6). Our per-round false-reject measurement (Section 5.3) is not new (it is the per-step tracking of Egashira et al. and TinyV, applied to offline SFT and to a defect of our own pipeline), and we claim neither it nor the patch control as a contribution. Supplementary Appendix A gives the full comparison.

3 Setup

3.1 The GVT Loop

Each round t has four steps. *Generate:* the current model samples k solutions per problem. *Verify:* `math-verify` extracts the final answer and compares it with the gold. *Select:* accepted solutions are kept. *Train:* a *fresh* LoRA adapter is attached to the *base* model and trained on the kept solutions for one epoch. Round 0 is the untrained base model; the round-t model ($t \geq 1$) is the base plus an adapter trained on the pool accepted at round $t-1$. Adapters are not accumulated, so earlier rounds affect later ones only through the data they generate; conclusions about (non-)compounding below apply to this restart-from-base design, which gives compounding less room than continued training would.

3.2 The Verifier and Its Directional Bias

A format probe that re-renders gold answers in individually valid forms shows that `\dfrac`/`\tfrac` are rejected 100% of the time on both datasets (the command is not parsed). On real solutions (one public Qwen2.5-7B-Instruct solution per problem for 200 MATH-500 problems, judged correct by a scripted normalised comparison) the false-negative rate is 7.8% (10 of 128; Wilson interval 4.3–13.8%) on MATH-500 and 0% on GSM8K. Eight of the ten fail when the whole solution is verified but pass when only the boxed answer is; most are compound answers, so they are probably largely the gold-storage defect described

below rather than extraction of a wrong token. The audit inspected only rejections, so it gives no false-positive rate. The verifier also cannot parse 51 of the 500 stored MATH-500 golds (10.2%); under V these problems always count as unsolved and can never enter the training pool (V' makes 46 of them parseable for boxed predictions). The offset cancels in within-run differences only if those problems would not otherwise change, which we did not check, and not in ratios or levels across partitions (11–18 such problems per exam partition).

The patch. For the control we rewrote `\dfrac` and `\tfrac` to `\frac` before parsing, removing the 100%-rejection class and nothing else.

The gold-storage defect (post hoc). Our pipeline stores the gold without a delimiter, and `math-verify` applies its full LaTeX extraction only inside `\boxed{}` or `$...$`, so a correct boxed prediction can fail against a bare gold (e.g. the prediction `\boxed{\frac{4}{3}}` against the stored gold `\frac43`). Re-scoring the saved generations with a verifier V' that wraps the gold in `\boxed{}` when the prediction contains one shows that this defect rejects 15.5–18.3% of the generations V' accepts. It would enlarge the round-0 kept pool by about a fifth to a quarter (440 → 542, +23%; this pool covers only the 350 selection problems, so its 18.8% rejected share differs from the 15.5–18.3% over all 500), against 0–4 examples for the `\dfrac` patch, and it lowers absolute pass@1 by 4.7–8.0 points in self-training. It is bound to the problem, not to the answer: on the 54–62 problems where it ever fires, it rejects 94–98% of correct answers in every round (87–98% where V parses the gold), and 29–34 of these problems have a gold the original verifier cannot parse at all, so no answer form avoids it there. Re-scored, sample accuracy still falls in all five runs we re-scored (three self-training, two train-on-gold); the original-verifier changes are 0.73–0.85 of the re-scored ones, an observed ratio that we do not use to rescale other numbers or intervals. All tables keep the original verifier, the one the loop trained with. Section 5.3 uses this defect for the adaptation audit; Supplementary Appendix B gives the full analysis.

3.3 Two Experimental Designs

Design P (pooled) loads both datasets (500 problems each) together and trains *one* adapter per round on the union of accepted solutions, with selection and evaluation on the same 1,000 problems. *Design I (isolated)* uses one dataset and one adapter per run, and holds out 30% of the 500 problems (150) as an `exam` set; generation runs on all 500 problems each round, but only the other 350 can enter the training pool, and all Design I pass rates are on the exam set. Table S1 summarises them.

Design I has three runs, all Qwen2.5-1.5B-Instruct, five rounds, $k = 8$, differing in one respect each: **Run A** (MATH-500, original verifier), **Run B** (MATH-500, patched verifier) and **Run C** (GSM8K, original verifier). Runs A and B were repeated at a second seed; since the seed also sets the train/exam partition, the repetition draws a different 150-problem exam set. Most of the later evidence comes from higher-rate runs introduced in Section 4.4; Table S4 (Supplementary Appendix E) lists every run. A "seed" sets the exam partition, the selection set

and the training randomness at once; runs with the same seed share the partition. "Five rounds" means round 0 (the untrained model) and four adapters, each trained once on the previous round's pool; "three rounds" means round 0 and two adapters, so the round-2 model is a single adapter (about 195 optimizer steps for Run A at 5×10^{-4}) trained on the round-1 pool. All runs use one model family, Qwen2.5.

3.4 Models, Data, Training, Metrics

We use Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct (Design P uses both, Design I the 1.5B model only) and the first 500 problems of the GSM8K test split [2] and MATH-500, the 500-problem subset of MATH [5]. The prompt is a single user turn in the model's chat template: "Solve the math problem. Put the final answer in `\boxed{}`." followed by the problem. Generation samples at temperature 0.8 with a cap of 512 new tokens; the other decoding settings come from the checkpoint's generation configuration (top-p 0.8, top-k 20, repetition penalty 1.1). Training is LoRA SFT (`transformers` 5.0.0, `peft` 0.19.1, TRL 1.10.0 `SFTTrainer`; rank 16, $\alpha = 32$, on `q_proj` and `v_proj`, LoRA dropout 0, weight decay 0, gradient clipping 1.0 and trainer seed 42, all library defaults read from the pinned sources; batch size 2, no accumulation; one epoch; AdamW with linear learning-rate decay and no warmup; Python and torch seeds set from the run seed; loss over the whole chat sequence, prompt included; fp16 on one Kaggle T4), with a training sequence limit of 512 tokens that includes the prompt. Because this limit equals the generation cap but also holds the prompt, any kept solution longer than about 512 tokens minus the prompt length is cut in training and loses its final answer and end-of-sequence token (24–26% of the round-0 pool; Section 4.5). All kept solutions are used, so the number of optimizer steps per round is half the pool size. The learning rate is 10^{-5} unless stated otherwise; this value was the default of our training code and was not tuned (Section 4.4). Metrics per round, on the exam set: pass@1 is the correctness of the *first* of the k samples (one draw per problem, so it carries sampling noise); pass@k; *sample accuracy*, the mean over all k samples; style counts over all generations (boxed, dfrac, dollar, ...); and per-problem added/lost counts between consecutive rounds, where a problem is *solved* in a round if at least one of its k samples is accepted. Added minus lost therefore equals $150 \times \Delta$pass@k, so these counts add only the gross churn behind the net change, not an independent test. pass@1 was fixed in advance; sample accuracy was not among the preset measures, but it estimates the same quantity from all k samples with less variance, so after the runs we chose it as the primary accuracy measure. Claims that rest on it, including the 2.2-point bound, are therefore post hoc in the choice of measure. "Correct" always means accepted by the verifier (preset `reward`), the same verifier that selects the training data.

Uncertainty. For every within-run change (round 0 against the last round) and every paired comparison of two runs on the same exam partition we report a 95% interval from a paired problem-level bootstrap (10,000 resamples of the

150 exam problems; 2,000 for the matched and audit columns), which respects the clustering of the k samples within a problem. These intervals do *not* cover training-seed variance, which we see only across two or three seeds: the replicate pair we use as a gauge ends 3.2 points of sample accuracy apart (a seed also changes the exam partition, so such pairs mix training and partition variance). We make no multiplicity correction and the runs are not independent (they share exam partitions and round-0 draws), so patterns across runs are descriptive, not tests; we apply the same reading to declines and to gains. Single proportions use Wilson intervals. *Detectable effects.* From the interval half-widths h in Table 2, a change would be detected with 80% power at the 5% level only if it exceeded about $2.8\,h/1.96$: about 10 points of pass@1, 8–10 of pass@k and 3.5–5.5 of sample accuracy on MATH-500. Because h omits training-seed variance (the replicate pair differs by 3.2 points and B minus A flips from +5.3 to −7.3 on near-identical pools, Section 4.4), these are optimistic thresholds, not bounds: even with h as the whole error, a gain at that size is missed 20% of the time, and probably more often (Supplementary Appendix C).

4 Results

4.1 Design P: An Apparent Collapse

In the pooled design the 1.5B model declines severely and with acceleration (pass@1 50.0% → 18.6%, pass@8 72.6% → 48.6%; Table S3, Supplementary Appendix D), on both datasets (Table S2); the 0.5B decline is small. The accepted pool contracts every round and `\boxed{}` falls from 5,608 to 2,693 of 8,000 generations. This design cannot be read causally: one adapter trains on both datasets' pools, it scores on the problems it selects from, and it trains on about 4.5× as many examples per round as the isolated MATH-500 run (3,962 against 872 at round 0); no run separates these (Supplementary Appendix D).

4.2 Design I at 10^{-5}: Small Declines, a Larger One on GSM8K

On the held-out exam (Fig. S1 and Table S5 in Supplementary Appendix E) pass@8 ends within 4 points of round 0 in every run. On MATH-500, pass@1 is 2.7–4.0 points lower at round 4 (run-log values, see Table 2), with every interval including zero (Table 2); sample accuracy falls 1.2–3.2 points, with an interval excluding zero only for B at seed 0 (A at seed 0 reaches 0.0), and matched finished accuracy 1.1–4.2, with intervals that only just exclude zero for A and B at seed 0, which are close to one replicate. We read this as a possible small decline, not an established one; what we do not see is a gain of the size the design could detect. Run C (GSM8K) declines more: pass@1 falls 12.7 points ($[-21.3, -4.0]$), almost all in finished accuracy, but it took about twice Run A's optimizer steps, so we do not attribute this to the dataset alone. Length and cap-hit share barely change on MATH-500, weak evidence that the adapter did little (Supplementary Appendix E).

4.3 The Patch Control Could Not Detect an Effect

Runs A and B differ only in the `\dfrac/\tfrac` patch. At round 4, B minus A is -0.7 [-3.3, $+2.0$] points of pass@1 at seed 0 and $+0.0$ [-2.7, $+2.7$] at seed 1 (recomputed from the saved generations; Table 2 lists run-log values, -0.7 at seed 1). This null is uninformative: `\dfrac` appears in 0.50–0.73% of generations, so the kept pools differ by at most 4 examples (under 0.5% of the 872-example pool), and at 10^{-5} the adapter changes the outputs little anyway, so a null is what prevalence alone predicts; the frequent gold-storage defect is in both arms.

4.4 Higher Learning Rates

Because 10^{-5} changed the MATH-500 outputs little, we reran Run A at 10^{-4} (three seeds), 2×10^{-4} (two) and 5×10^{-4} (two, with Run B), with three rounds and $k = 4$ (Table 1). These runs differ from the 10^{-5} runs in k, rounds, pool size and the pass@k metric as well as rate, so only 10^{-4} to 5×10^{-4} isolates the rate. *At* 5×10^{-4} *every A/B run declines:* pass@1 falls 4.0–11.3 points and the sample-accuracy intervals exclude zero in all four A/B runs, though same-seed A and B are closer to two replicates than four. *The decline may grow with the rate on one seed* (sample-accuracy changes at 10^{-5} and 10^{-4} are similar, -1.2 to -3.2 and -1.0 to -3.7): at seed 1 sample accuracy falls 1.5, 3.5 and 8.3 points across the three rates, at seed 2 an almost flat 3.7, 3.8 and 4.7. The cap-hit share rises only at 5×10^{-4}. *The patch has no consistent effect:* B minus A in round-2 pass@1 is $+5.3$ at seed 1 and -7.3 [-14.0, -0.7] at seed 2, a sign flip from pools differing by 0–3 examples: trajectory noise. *Style does not separate imitation from drift:* boxed share falls with the pool, which both would produce, and the no-filter and gold arms decline with no verifier in the selection step (they are still scored by it), so generic SFT damage fits as well.

Table 1. Run A (MATH-500, original verifier, $k = 4$, three rounds) across learning rates and seeds: pass@1, pass@4 and sample accuracy (600 exam generations) at round 0 $\rightarrow$ round 2, the share of round-2 generations that hit the 512-token cap (decoded text re-tokenised to ≥ 508 tokens; round 0 is 43.0–43.2%, 48.2% at seed 3), and problems added/lost summed over rounds 1–2. The two 5×10^{-4} rows are the A rows of Table S6.

Rate	Seed	pass@1	pass@4	Sample acc.	Cap-hit (R2)	Added/lost
10^{-4}	1	33.3 $\rightarrow$ 34.0	50.0 $\rightarrow$ 49.3	33.7 $\rightarrow$ 32.2	42.7%	20/21
10^{-4}	2	31.3 $\rightarrow$ 28.0	51.3 $\rightarrow$ 45.3	30.8 $\rightarrow$ 27.2	42.8%	25/34
10^{-4}	3	29.3 $\rightarrow$ 26.7	40.0 $\rightarrow$ 41.3	28.5 $\rightarrow$ 27.5	45.7%	16/14
2×10^{-4}	1	33.3 $\rightarrow$ 31.3	50.0 $\rightarrow$ 46.7	33.7 $\rightarrow$ 30.2	40.8%	23/28
2×10^{-4}	2	31.3 $\rightarrow$ 25.3	51.3 $\rightarrow$ 45.3	30.8 $\rightarrow$ 27.0	39.7%	21/30
5×10^{-4}	1	33.3 $\rightarrow$ 23.3	50.0 $\rightarrow$ 42.7	33.7 $\rightarrow$ 25.3	46.2%	22/33
5×10^{-4}	2	31.3 $\rightarrow$ 27.3	51.3 $\rightarrow$ 44.7	30.8 $\rightarrow$ 26.2	51.2%	26/36

Table 2. Change from round 0 to the last round (percentage points) with 95% paired problem-level bootstrap intervals over the 150 exam problems (10,000 resamples; `bootstrap_ci.py` in the code repository). Last round is R4 at 10^{-5} ($k = 8$) and R2 elsewhere ($k = 4$). "Finished, matched" is the last-round accuracy of generations below the 512-token cap minus the untrained model's accuracy on the same problem mix (Section 4.5; 2,000 resamples, `capanat.py` in the repository). The intervals do not include training-seed variance. Recomputed values match the reported ones except six round values that differ by one problem (Supplementary Appendix C), e.g. B, seed 1 at 10^{-5}, whose round-4 pass@8 is 58.0% against 58.7%; the matched column uses the separate re-scoring described there.

Run, seed	Δpass@1	Δpass@k	Δsample acc.	Finished, matched
Learning rate 10^{-5}				
A, 0	-2.7 $[-9.3, 4.0]$	-1.3 $[-8.0, 5.3]$	-2.4 $[-4.8, 0.0]$	-3.8 $[-7.2, -0.5]$
A, 1	-3.3 $[-10.0, 3.3]$	$+0.0$ $[-6.0, 6.0]$	-1.5 $[-4.1, 1.0]$	-2.3 $[-5.9, 1.2]$
B, 0	-3.3 $[-10.0, 3.3]$	-4.0 $[-10.7, 2.7]$	-3.2 $[-5.9, -0.6]$	-4.2 $[-7.8, -0.9]$
B, 1	-4.0 $[-10.7, 2.7]$	-1.3 $[-7.3, 4.7]$	-1.2 $[-3.7, 1.3]$	-1.1 $[-4.6, 2.6]$
C, 0	-12.7 $[-21.3, -4.0]$	-2.7 $[-7.3, 2.0]$	-13.1 $[-16.7, -9.8]$	-12.8 $[-16.5, -9.4]$
Learning rate 10^{-4}				
A, 1	$+0.7$ $[-6.7, 8.0]$	-0.7 $[-6.0, 4.7]$	-1.5 $[-5.3, 2.2]$	-4.3 $[-9.6, 1.1]$
A, 2	-3.3 $[-9.3, 2.7]$	-6.0 $[-12.7, 0.7]$	-3.7 $[-7.3, -0.2]$	-3.5 $[-9.2, 2.3]$
A, 3	-2.7 $[-10.0, 4.0]$	$+1.3$ $[-4.7, 7.3]$	-1.0 $[-4.3, 2.2]$	-0.6 $[-6.0, 4.7]$
Learning rate 2×10^{-4}				
A, 1	-2.0 $[-8.0, 4.0]$	-3.3 $[-10.0, 2.7]$	-3.5 $[-7.2, 0.0]$	-4.0 $[-8.9, 0.5]$
A, 2	-6.0 $[-12.7, 0.7]$	-6.0 $[-12.7, 0.7]$	-3.8 $[-7.5, -0.2]$	-4.7 $[-10.1, 1.0]$
Learning rate 5×10^{-4}				
A, 1	-10.0 $[-16.7, -3.3]$	-7.3 $[-14.0, 0.0]$	-8.3 $[-12.3, -4.5]$	-9.1 $[-14.2, -4.2]$
A, 2	-4.0 $[-11.3, 3.3]$	-6.7 $[-13.3, 0.0]$	-4.7 $[-8.2, -1.2]$	-5.5 $[-11.7, 0.1]$
B, 1	-4.7 $[-12.0, 2.7]$	-2.0 $[-8.7, 4.7]$	-6.7 $[-10.5, -2.8]$	-6.8 $[-12.4, -1.0]$
B, 2	-11.3 $[-17.3, -5.3]$	-11.3 $[-18.0, -4.7]$	-9.2 $[-12.8, -5.5]$	-5.9 $[-10.9, -1.0]$
No filter, 1	-6.0 $[-12.7, 1.3]$	-9.3 $[-16.7, -2.7]$	-8.2 $[-12.2, -4.3]$	-10.6 $[-16.4, -5.5]$
No filter, 2	-8.7 $[-16.0, -1.3]$	-12.0 $[-19.3, -4.7]$	-10.2 $[-14.2, -6.3]$	-6.6 $[-11.4, -2.0]$
Gold, 1	-18.0 $[-26.7, -10.0]$	-20.7 $[-28.7, -12.7]$	-19.7 $[-25.3, -14.2]$	-25.6 $[-32.1, -18.9]$
Gold, 2	-21.3 $[-28.7, -14.0]$	-20.7 $[-29.3, -12.7]$	-18.7 $[-23.7, -13.7]$	-25.9 $[-31.7, -20.0]$
C, 1	-22.0 $[-29.3, -15.3]$	-9.3 $[-15.3, -3.3]$	-15.3 $[-19.8, -11.2]$	-12.1 $[-16.6, -7.9]$

Two controls at 5×10^{-4}. The *no-filter* run trains on all k samples of every selection problem (1,400 examples per round), removing the verifier; the *train-on-gold* run trains on the 350 reference solutions every round (Table 3). Neither is a single-variable control: no-filter takes 3–4× Run A's optimizer steps, train-on-gold about the same. Removing the filter is not better than keeping it beyond noise and is lower on pass@4 at both seeds (round-2 differences -2.0 and -5.3 points, intervals including zero), but the dose difference means this cannot be attributed to the filter; the no-filter pool is also truncated far more often in training (62.4% against 24.5% at round 0). Training on reference solutions is followed by a much larger decline (pass@4 down about 21 points at both seeds) while the model adopts the reference style (`$`-delimited spans 5% → 98%). Post hoc checks bound verifier error from below only (about 2 points of sample accuracy among boxed answers), and the 14% of round-2 generations without `\boxed` are not covered, so we think this fall is unlikely to be mainly verifier error but cannot bound it from above. That no arm, this one included, improves on the untrained model means we have no positive reference: the nulls above may reflect

the recipe (LoRA on q/v projections, loss over the prompt, a 512-token limit) rather than self-training. Run C at 5×10^{-4} (GSM8K, one run) falls 22.0 points of pass@1, from both less accurate and more unfinished solutions (Supplementary Appendix G).

Table 3. Controls at 5×10^{-4} (MATH-500, $k = 4$, three rounds), on the exam partition of Run A at the same seed. Pool is the kept pool at round 0 → round 2; the round-2 model trains on the round-1 pool (392 and 386 examples, about 195 steps) and optimizer steps are half the pool size.

Run	Pool	pass@1			pass@4		
		R0	R1	R2	R0	R1	R2
Seed 1							
A (verifier-filtered)	440→326	33.3	32.7	23.3	50.0	49.3	42.7
No filter	1400	33.3	26.7	27.3	50.0	49.3	40.7
Train on gold	350	33.3	18.0	15.3	50.0	33.3	29.3
Seed 2							
A (verifier-filtered)	424→350	31.3	31.3	27.3	51.3	46.0	44.7
No filter	1400	31.3	34.7	22.7	51.3	48.7	39.3
Train on gold	350	31.3	8.7	10.0	51.3	28.7	30.7

4.5 Anatomy of the Decline

This subsection is post hoc and descriptive. Sample accuracy mixes whether a finished solution is right with whether the model finishes within the 512-token cap at all. Re-tokenising the saved text, we call a generation *cap-hit* at ≥ 508 tokens. At the rates where the share does not rise, 40–46% of round-2 MATH-500 generations already hit the cap (Table 1), so the cap binds from the start. At 5×10^{-4}, compared with the untrained model on the same problem mix, finished generations lose 5.5–10.6 points, with intervals excluding zero in five of six runs (the six include two same-seed A/B pairs, which are close to replicates), while the cap-hit share rises 2–14 points. The matched measure conditions on which problems finish after training, so it is descriptive. A three-part decomposition attributes 12–30% of the decline to the cap share in the three original seed-1 runs (and −2 and 5% in two further runs) and 52–72% at seed 2 (Table S9, Supplementary Appendix H). At 10^{-4} and 2×10^{-4} the cap-hit share does not rise and finished accuracy carries the whole change. Train-on-gold is different: responses shorten by more than half and finished accuracy collapses. Retraining without truncated examples, or re-scoring at a 1,536-token cap, leaves the deficit in place (Supplementary Appendix H).

4.6 A Final Round with a Revised Recipe

Because no earlier arm improved, we ran one last round with several recipe changes made at once, aimed at the suspected causes (no ablation of any single one): loss on the completion only, LoRA on all linear layers, a 1,024-token cap and training limit, normalised gold answers, and a pool of 300 problems from the MATH training split, so that the exam is all 500 MATH-500 problems. It is 10^{-4} with $k = 2$ and a single adapter (round 0 to round 1); the loop has three seeds, the gate one. We report no overlap check of the 300 pool problems against MATH-500. The verifier-filtered loop (298 examples kept) is compared with training on the 300 reference solutions, a positive gate that must raise accuracy for the recipe to count as able to help.

Table 4. Final round, MATH-500 (500 exam problems, $k = 2$, 10^{-4}). Change from each run's own untrained round 0 to round 1 in points (round-0 pass@1 is 49.2, 46.8, 48.2 at seeds 1–3, from sampling), with 95% paired problem-level bootstrap intervals that omit training-seed variance. The gold arm has one seed.

Arm	pass@1	pass@2	Sample acc.
Untrained (seed 1)	49.2	59.4	–
Loop, seed 1	-4.8 $[-9.0, -0.6]$	-3.0 $[-6.8, +0.6]$	-2.9 $[-5.8, 0.0]$
Loop, seed 2	-3.2 $[-6.8, +0.6]$	-6.2 $[-9.8, -2.4]$	-5.3 $[-8.1, -2.4]$
Loop, seed 3	-4.2 $[-8.4, -0.2]$	-5.6 $[-9.4, -1.6]$	-5.4 $[-8.3, -2.4]$
Loop, mean of seeds	-4.1	-4.9	-4.5
Train on gold	-32.2 $[-36.8, -27.6]$	-32.0 $[-36.6, -27.2]$	-30.2 $[-33.8, -26.6]$

The gate fails. Training on reference solutions lowers pass@1 from 49.2% to 17.0%, and the model adopts the reference style ($-delimited spans in 60 of 1,600 generations at round 0, 1,577 at round 1, counted over the exam and selection problems), as in the earlier train-on-gold arms. A rerun at 2×10^{-5} fails the same way (49.2% $\rightarrow$ 20.6%, $-spans in 1,558 of 1,600 generations), so the failure is not specific to the rate. We noticed the shift to the reference style but did not test it as the cause. That style carries no `\boxed` answer, which both graders read, and a collapse of this size from 300 reference solutions is also what a pipeline defect would produce. The gate therefore fails to validate the revised recipe, and the recipe cannot be called corrected. We have no recipe known to raise accuracy, so the loop's decline cannot be attributed to self-training or to the verifier. The verifier-filtered loop loses pass@1 in all three seeds (4.8, 3.2, 4.2 points; two intervals exclude zero) and sample accuracy in all three (intervals exclude zero for seeds 2 and 3). The mean pass@1 change over the three seeds is -4.1 (t interval $[-6.1, -2.1]$), and -4.5 for sample accuracy ($[-8.1, -1.0]$). The loop does not help; the point estimates are negative in all three seeds, but the sample is three seeds with one exam draw and one rate, so we do not claim that it costs accuracy, and we did not isolate the cause. The detectable effect is about 6 points of pass@1 ($2.8h/1.96$), against about 10 before.

Injected bias. As intended positive controls for the audit (neither turns out to be one) we reran the loop once each (seed 1, 10^{-4}) with a selection verifier that rejects `\boxed` answers (exam verifier clean). In the first, a hash-chosen 30% of problems are affected: the `\boxed` share falls 4.9 points on them and 3.1 on the others (difference -1.8, $[-7.4, 3.8]$); a hash cannot be learned from the problem, so this control could not elicit selective avoidance. In the second the rule is visible: every `\boxed` answer is rejected when the gold contains `\frac` (49 of the 300 selection problems, 263 examples kept instead of 298; 70 of the 500 exam problems, where the shares below are measured). The share on those problems stays at 95.0% and on the others goes from 90.2% to 87.4% (difference $+2.8$, $[-2.8, +8.3]$), against -3.7, -2.7 and -5.5 for the same split in the three unbiased seeds, a shift in the direction opposite to avoidance; pass@1 falls 7.0 points ($[-11.2, -3.0]$), more than in the unbiased seeds, which we do not interpret (one run). In neither run does the `\boxed` share fall selectively where boxing is penalised, so neither validates the audit.

5 Discussion

5.1 Answer to the Research Question

Does the model get better at mathematics? Not detectably. No run raises pass@1 or pass@k with an interval excluding zero, at any rate; in the 500-problem final round pass@1 falls 3.2–4.8 points in each of three seeds. The sample-accuracy figure of about 2.2 points is the largest upper limit among observed intervals, a post hoc figure that omits seed variance, not a power statement; on the pre-set measures the design could detect only about 10 points of pass@1 and 8–10 of pass@k (6 and 5 in the final round). Because no arm improves and the positive gate failed, the null belongs to this recipe, and the study cannot say whether self-training in general helps. Every score except the final-round check below is graded by the verifier that selects the data (re-scoring five runs, three self-training and two train-on-gold, with V' leaves the decline in place, Section 3). On the final-round generations a second grader (normalised string match of the last `\boxed` against the gold, no `math-verify`; `indep_grade.py`) gives pass@1 changes of -3.6, -2.2, -4.6 for the three seeds (-30.0 for the gold arm) and sample-accuracy changes of -2.5, -5.5, -5.3 (same sign and size). It reads only the last `\boxed`, so a falling boxed share lowers both graders, and it does not rule out a format artefact in the gate. Earlier runs are not independently graded.

Does it learn to write what the verifier accepts? The data cannot settle this. The pre-set pattern requires pass@1 to rise, which never happens detectably (largest rise $+0.7$ $[-6.7, +8.0]$), so its silence is no evidence either way; no run varies the frequent gold-storage defect, and the audit of Section 5.3 is too weak.

5.2 Interpretation

One explanation of the Design P curves is a self-reinforcing shrinking pool shaped by the biased verifier. The data neither support nor rule it out: the patch tests

a rare bias, and the no-filter control declines without any selection verifier but with more dose (and, with a constant pool, cannot test the shrinking pool). At this resolution, *removing a rare format defect, or the filter entirely, made no consistent difference* (no filter is lower on pass@k at 3–4× the dose); neither tests the systematic false positives Egashira et al. find harmful in RLVR.

5.3 A Proposed Audit for GVT with a Biased Verifier

Our pre-set rule tied "learning the verifier's style" to rising accuracy, so it could not speak once accuracy fell. From post hoc analyses we propose four checks for self-training with a rule-based verifier, as a protocol for future studies, not as tested methods.

(i) A per-round false-reject audit. Fix a repaired verifier V' that corrects a known defect class of the training verifier V. At each round t, on the loop's own generations, compute

$$r_t = \frac{\#\{\text{accepted by } V' \text{ but rejected by } V\}}{\#\{\text{accepted by } V'\}}.$$

If the model learns to write what V accepts, r_t may fall over rounds, but it also falls when fewer answers are boxed: since V' differs from V only on boxed predictions, $r_t = r_t^{\text{boxed}} \times$ (boxed share among V'-accepted) holds as an identity. This is the per-step tracking of RLVR studies [4,16] applied to offline SFT (Table 5; `rt_audit.py`). For the gold-storage defect it falls in all three self-training runs we re-scored, by 2.6, 2.6 and 1.5 points, with intervals reaching or just touching zero. The fall is accounted for by the falling `\boxed` share among correct answers ($91 \to 76\%$ and $93 \to 78\%$ at 5×10^{-4}): an identity, not a test, which does not exclude adaptation. On boxed answers alone the changes (-0.4 to $+0.7$ points) have intervals (half-widths 1.6–3.6) that contain the overall fall of the same run (-2.6, -2.6, -1.5), so "flat" means not detected. The defect is bound to the problem: it fires on 54–62 problems, rejecting 94–98% of correct answers, and 29–34 of them have an unparseable gold, so no answer form escapes it. On the 24–28 affected problems where some form could, r_t moves by -5.5, -1.6 and -0.2 points (intervals 10–26 wide), and the pool holds only 2–12 such unboxed examples per round, so we expect weak pressure to adapt (untested; no dose–response). The two injected-bias runs (Section 4.6) show no selective avoidance, but neither can tell us whether the audit would detect it. Train-on-gold, with no verifier in the loop, gives r_t rises of 2.1–3.3 points (2.8 and 1.7 on boxed answers alone; Table 5), so r_t drifts as much as the falls we observe, unrelated to the verifier. It should be reported on the answers and problems where V and V' can differ and an accepted form exists, with the count of training examples that could teach avoidance.

(ii) Decompose declines under a fixed cap. Report the change in sample accuracy split into a finished-accuracy part and a cap-share part (Table S9, Supplementary Appendix H). Here the split is descriptive and its balance differs by partition: at 5×10^{-4} the cap share accounts for 12–30% of the decline in the three original seed-1 runs and 52–72% at seed 2.

Table 5. The false-reject audit for the gold-storage defect (post hoc), in percent (all 500 problems, selection problems included), round 0 → last round, with the change and a 95% problem-level bootstrap interval (2,000 resamples). "Boxed only" restricts to predictions containing `\boxed`; "affected, parseable" to problems where the defect ever fires and V can parse the gold. Pool: examples per round from affected problems that V accepts (and so avoid the defect).

Run	r_t subset	R0	Last	Δ	95% interval	Pool
A, 5×10^{-4}, seed 1	all	18.3	15.7	−2.6	$[-5.3, 0.3]$	3–5
	boxed only	20.0	20.7	+0.7	$[-2.9, 4.3]$	
	affected, parseable	92.8	87.2	−5.5	$[-20.1, 6.3]$	
A, 5×10^{-4}, seed 2	all	18.1	15.5	−2.6	$[-5.2, 0.1]$	2–3
	boxed only	19.4	19.8	+0.4	$[-2.8, 3.4]$	
	affected, parseable	95.1	93.5	−1.6	$[-14.7, 9.5]$	
A, 10^{-5}, seed 0	all	18.3	16.8	−1.5	$[-2.9, 0.0]$	9–12
	boxed only	19.9	19.5	−0.4	$[-2.0, 1.2]$	
	affected, parseable	92.3	92.0	−0.2	$[-4.9, 4.6]$	
Gold, seed 1	all	18.2	21.5	+3.3	$[-1.2, 8.1]$	—
	boxed only	20.0	22.8	+2.8	$[-2.0, 7.7]$	
	affected, parseable	95.5	94.7	−0.7	$[-9.1, 6.2]$	
Gold, seed 2	all	18.0	20.2	+2.1	$[-2.2, 6.7]$	—
	boxed only	19.4	21.1	+1.7	$[-3.3, 6.5]$	
	affected, parseable	95.1	97.6	+2.5	$[-4.8, 11.9]$	

(iii) Check the dose of a control first (our `\dfrac` patch changed 0–4 examples; the unvaried defect a fifth to a quarter of the pool) and *(iv) match dose across designs*, which the pooled design did not.

6 Limitations

Unequal training doses confound every cross-arm comparison, and the controls are underpowered: the patch changes 0–4 training examples, and the no-filter and train-on-gold controls have two seeds each. *Intervals cover only problem draw and sampling,* not training-seed variance; the final-round loop has three seeds, the gate and the bias controls one, and no multiplicity correction is applied. *The learning-rate grid is coarse* and changes k, rounds and dose with the rate. *Power is limited* (about 10 points of pass@1 with 150 exam problems, about 6 in Section 4.6). *The adaptation audit is weak:* 24–28 problems, 2–12 avoidance examples per round, and no valid positive control. *The final-round loop has no no-filter arm. The verifier audit* used other models' solutions and inspected only rejections; the gold-storage defect ran unvaried through every earlier run, whose headline numbers are graded by the biased verifier that also selects the data, and the final-round string-match grader shares the `\boxed` dependence. The final-round decline and the gate failure cannot be attributed to

any single recipe change. *Truncation:* the 512-token cap binds from the start (a quarter of the pool is cut). *Scope:* one model family and size, all instruction-tuned (no base model), one verifier, a restart-from-base design that leaves compounding little room (no continued training), possible Qwen2.5 contamination on MATH/GSM8K (unquantified), and a default repetition penalty (1.1) that may depress the baseline.

7 Conclusion

We asked whether generate–verify–select–train iterations make a 1.5B model better at mathematics or only teach it to write what a rule-based verifier accepts. The design detected no pass@1 or pass@k gain at any rate (it could see only gains of about 10 and 8–10 points; 6 in a final 500-problem round, where pass@1 falls 3.2–4.8 points in all three seeds). Because training on reference solutions lowers pass@1 by about 30 points even with a revised recipe, this is an inconclusive null for the recipe and says little about self-training, and since scores are graded by the selecting verifier (apart from one string-match check), neither branch of the question is answered. The post hoc false-reject audit is uninformative about adaptation: its fall is tied to the falling `\boxed` share by an identity, and it has little power (about 25 affected problems, a handful of avoidance examples per round, no valid positive control). At the highest rate the model is less accurate on the solutions it finishes. The pooled design's decline is absent in the isolated runs at 10^{-5}, which use about 4.5× fewer examples, so dose and pooling are confounded.

A dose-matched control for the gold-storage defect, a recipe that passes the positive gate, and a second grader come first (Supplementary Appendix K).

Acknowledgments. This work was carried out as an undergraduate research project. No external funding was received.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

Code availability Code, saved generations and analysis scripts are at https://github.com/tsan2711/DACNTT_2026 (tag `paper-2026-11b`). Generation-time token counts, Design P's training logs and the 1,536-token re-scoring outputs were not saved.

Use of AI tools The authors used Claude (Anthropic), through Claude Code, to help write the manuscript, the analysis and training scripts, and to run automated reviews. The authors designed the study, ran the experiments and take responsibility for the content.

References

1. Cai, X.-Q., Wang, W., Liu, F., Liu, T., Niu, G., Sugiyama, M.: Reinforcement Learning with Verifiable yet Noisy Rewards under Imperfect Verifiers. arXiv:2510.00915 (2025)

2. Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., Schulman, J.: Training Verifiers to Solve Math Word Problems. arXiv:2110.14168 (2021)
3. Dang, Q.-A., Ngo, C.: Reinforcement Learning for Reasoning in Small LLMs: What Works and What Doesn't. arXiv:2503.16219 (2025)
4. Egashira, K., Vero, M., Dekoninck, J., Dörner, F.E., Staab, R., Vechev, M.: Delay, Plateau, or Collapse: Evaluating the Impact of Systematic Verification Error on RLVR. arXiv:2605.02909 (2026)
5. Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D., Steinhardt, J.: Measuring Mathematical Problem Solving With the MATH Dataset. arXiv:2103.03874 (2021)
6. Huang, Y., Zeng, W., Zeng, X., Zhu, Q., He, J.: From Accuracy to Robustness: A Study of Rule- and Model-based Verifiers in Mathematical Reasoning. arXiv:2505.22203 (2025)
7. Kim, J., Luo, X., Kim, M., Lee, S., Kim, D., Jeon, J., Li, D., Yang, Y.: Why Does Self-Distillation (Sometimes) Degrade the Reasoning Capability of LLMs? arXiv:2603.24472 (2026)
8. Qiao, X., Du, X., Liu, W., Zhang, J., Mai, P., Zhang, M., Pang, Y.: When Sample Selection Bias Precipitates Model Collapse. arXiv:2606.13732 (2026)
9. Shao, R., Li, S.S., Xin, R., Geng, S., Wang, Y., Oh, S., Du, S.S., Lambert, N., Min, S., Krishna, R., Tsvetkov, Y., Hajishirzi, H., Koh, P.W., Zettlemoyer, L.: Spurious Rewards: Rethinking Training Signals in RLVR. arXiv:2506.10947 (2025)
10. Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., Gal, Y.: AI models collapse when trained on recursively generated data. Nature (2024). https://doi.org/10.1038/s41586-024-07566-y
11. Singh, A., Co-Reyes, J.D., Agarwal, R., et al.: Beyond Human Data: Scaling Self-Training for Problem-Solving with Language Models. arXiv:2312.06585 (2023)
12. Gulcehre, C., Le Paine, T., Srinivasan, S., et al.: Reinforced Self-Training (ReST) for Language Modeling. arXiv:2308.08998 (2023)
13. Strozzi, I.L.: Teacher-Free Self-Training Amplifies but Does Not Compound: A Pass@k Crossover on a Free-Verifier Domain. arXiv:2606.07856 (2026)
14. Wang, Y., Yang, Q., Zeng, Z., Ren, L., Liu, L., Peng, B., Cheng, H., He, X., Wang, K., Gao, J., Chen, W., Wang, S., Du, S.S., Shen, Y.: Reinforcement Learning for Reasoning in Large Language Models with One Training Example. In: NeurIPS (2025). arXiv:2504.20571
15. Xu, C., Yan, N., Chen, L., Kechadi, M.-T.: Phantom Gains: Auditing Self-Improvement Against a Measured Null. arXiv:2608.20290 (2026)
16. Xu, Z., Li, Y., Jiang, F., Ramasubramanian, B., Niu, L., Lin, B., Poovendran, R.: TinyV: Reducing False Negatives in Verification Improves RL for LLM Reasoning. arXiv:2505.14625 (2025)
17. Yuan, Z., Yuan, H., Li, C., Dong, G., Lu, K., Tan, C., Zhou, C., Zhou, J.: Scaling Relationship on Learning Mathematical Reasoning with Large Language Models. arXiv:2308.01825 (2023)
18. Yue, Y., Chen, Z., Lu, R., Zhao, A., Wang, Z., Yue, Y., Song, S., Huang, G.: Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model? arXiv:2504.13837 (2025)
19. Zelikman, E., Wu, Y., Mu, J., Goodman, N.D.: STaR: Bootstrapping Reasoning With Reasoning. arXiv:2203.14465 (2022)
20. Zeng, W., Huang, Y., Liu, Q., Liu, W., He, K., Ma, Z., He, J.: SimpleRL-Zoo: Investigating and Taming Zero Reinforcement Learning for Open Base Models in the Wild. arXiv:2503.18892 (2025)